Unitary Optimization Does Not Restore Superadditivity of Quantum Relative Entropy

Ao-Xiang Liu

(Dated: August 2, 2026)

A proposed repair of superadditivity for quantum relative entropy asks whether unitary rotations of a bipartite state and its marginals can make the inequality valid for every full-rank reference state. We disprove this conjecture in the smallest bipartite dimension. For $\rho_{AB} = \mathbb{I}_4/4$ and $\sigma_{AB} = \mathrm{diag}(1, 1, 1, 13)/16$, every allowed unitary leaves the first argument invariant, while the superadditivity gap is exactly $\frac{1}{4}\log_2(2401/3328) < 0$. The same example also refutes the proposed spectral inequality and the later local-unitary strengthening. More generally, every $m \times n$ system with $m, n \geq 2$ admits an explicit full-rank diagonal family whose optimized gap tends to minus infinity. The failures persist on open sets. The obstruction is therefore classical and robust: unitary-orbit optimization cannot correct the double counting of distinguishability generated by a correlated reference state.

Status. This document records an analytic research result that has not undergone external peer review. The calculations and quantifiers have been checked independently. A targeted literature and citation audit through August 2, 2026 did not locate a prior proof or counterexample to the conjectures treated below; this statement is not an absolute claim of priority.

I. PROBLEM AND RESULT

For finite-dimensional states ρ and σ, with σ full rank, the quantum relative entropy is

$$D(\rho\|\sigma) = \mathrm{Tr}[\rho(\log\rho - \log\sigma)]\,. \tag{1}$$

If the bipartite reference state is a product, $\sigma_{AB} = \sigma_A \otimes \sigma_B$, relative entropy satisfies the standard superadditivity inequality

$$D(\rho_{AB}\|\sigma_A \otimes \sigma_B) \geq D(\rho_A\|\sigma_A) + D(\rho_B\|\sigma_B). \tag{2}$$

The same statement fails for a correlated reference σ_{AB}. Zhang, Wu, and Fei proposed repairing it by optimizing the first argument over unitary orbits [1]: for every ρ_{AB} and every full-rank σ_{AB}, do there exist independent unitaries U_{AB}, U_A, and U_B such that

$$\begin{aligned} &D(U_{AB}\rho_{AB}U_{AB}^\dagger\|\sigma_{AB}) \\ &\quad\geq D(U_A\rho_A U_A^\dagger\|\sigma_A) + D(U_B\rho_B U_B^\dagger\|\sigma_B)? \end{aligned} \tag{3}$$

A subsequent study reported numerical support for all two-qubit pairs and also conjectured a stronger version in which the global rotation is the local unitary $U_A \otimes U_B$ appearing on the right-hand side [2]. These are Eq. (4.2) of Ref. [1], Eq. (1.1) and Conjecture 3.1 of Ref. [2], respectively.

We show that all these conjectures fail. The key observation is to choose the first argument maximally mixed. The unitary freedom then disappears both globally and locally, reducing the question to a determinant inequality. An explicit reference state reverses that inequality already for two commuting two-qubit states, and the same construction works in every nontrivial bipartite dimension.

II. FLAT INPUTS REDUCE THE PROBLEM TO DETERMINANTS

For a full-rank state ω on a d-dimensional Hilbert space, Eq. (1) gives

$$D\left(\frac{\mathbb{I}_d}{d}\middle\|\omega\right) = -\log d - \frac{1}{d}\log\det\omega. \tag{4}$$

Consequently, when $\rho_{AB} = \mathbb{I}_{mn}/(mn)$, the unitary conjecture is equivalent to

$$(\det\sigma_{AB})^{1/(mn)} \leq (\det\sigma_A)^{1/m}(\det\sigma_B)^{1/n}. \tag{5}$$

Indeed, every unitary in Eq. (3) fixes the corresponding first argument. There is no general determinant inequality of the form (5).

Theorem 1 (Counterexamples in every bipartite dimension)**.** *For every $m, n \geq 2$, there are full-rank commuting states ρ_{AB} and σ_{AB} on $\mathbb{C}^m \otimes \mathbb{C}^n$ for which Eq. (3) fails for all choices of U_{AB}, U_A, and U_B. The violation can be made arbitrarily large.*

Proof. In a fixed product basis, let

$$\begin{aligned} \rho_{AB} &= \frac{\mathbb{I}_{mn}}{mn}, \\ \sigma_{AB}^{(\varepsilon)} &= \varepsilon \sum_{(i,j)\neq(m,n)} |i,j\rangle\langle i,j| + h|m,n\rangle\langle m,n|, \end{aligned} \tag{6}$$

where $h = 1 - (mn-1)\varepsilon$ and $0 < \varepsilon < 1/(mn-1)$. The reference state is full rank. Define

$$a = 1 - n(m-1)\varepsilon, \qquad b = 1 - m(n-1)\varepsilon. \tag{7}$$

Taking partial traces in Eq. (6) gives the marginal spectra

$$\begin{aligned} \mathrm{spec}(\sigma_A^{(\varepsilon)}) &= \big(\underbrace{n\varepsilon, \ldots, n\varepsilon}_{m-1}, a\big), \\ \mathrm{spec}(\sigma_B^{(\varepsilon)}) &= \big(\underbrace{m\varepsilon, \ldots, m\varepsilon}_{n-1}, b\big). \end{aligned} \tag{8}$$

Both marginals are full rank throughout the stated range. Let the superadditivity gap be

$$\Delta_{m,n}(\varepsilon) = D(\rho_{AB}\|\sigma_{AB}^{(\varepsilon)}) - D(\rho_A\|\sigma_A^{(\varepsilon)}) - D(\rho_B\|\sigma_B^{(\varepsilon)}). \tag{9}$$

Using Eq. (4) and

$$\begin{aligned}\det \sigma_{AB}^{(\varepsilon)} &= \varepsilon^{mn-1} h,\\ \det \sigma_{A}^{(\varepsilon)} &= (n\varepsilon)^{m-1} a,\\ \det \sigma_{B}^{(\varepsilon)} &= (m\varepsilon)^{n-1} b,\end{aligned} \tag{10}$$

we obtain the exact expression

$$\Delta_{m,n}(\varepsilon) = \frac{1}{mn} \log\left[\frac{\varepsilon^{(m-1)(n-1)} n^{n(m-1)} m^{m(n-1)} a^n b^m}{h}\right]. \tag{11}$$

Because $(m-1)(n-1) > 0$, the quantity in brackets tends to zero as $\varepsilon \to 0^+$. Thus $\Delta_{m,n}(\varepsilon) \to -\infty$, so the gap is negative for every sufficiently small positive ε. All first arguments in Eq. (3) are proportional to identity operators; hence no unitary choice changes this conclusion. □

The theorem does more than supply a counterexample at a fixed dimension or parameter value. It gives a uniform construction for every $m, n \geq 2$, and its violation is unbounded as the full-rank reference approaches the boundary of state space.

III. AN EXACT TWO-QUBIT CERTIFICATE

The smallest member of the family has a particularly simple rational form. In the product-basis order $00, 01, 10, 11$, take

$$\rho_{AB} = \frac{\mathbb{I}_4}{4}, \qquad \sigma_{AB} = \frac{1}{16}\,\mathrm{diag}(1, 1, 1, 13). \tag{12}$$

Both states are full rank, and

$$\rho_A = \rho_B = \frac{\mathbb{I}_2}{2}, \qquad \sigma_A = \sigma_B = \frac{1}{8}\,\mathrm{diag}(1, 7). \tag{13}$$

Using logarithms to base two,

$$\begin{aligned} D(\rho_{AB}\|\sigma_{AB}) &= 2 - \frac{1}{4}\log_2 13,\\ D(\rho_A\|\sigma_A) = D(\rho_B\|\sigma_B) &= 2 - \frac{1}{2}\log_2 7. \end{aligned} \tag{14}$$

Therefore

$$\Delta_{2,2}\left(\frac{1}{16}\right) = \frac{1}{4}\log_2\left(\frac{2401}{3328}\right) = -0.1177550074\ldots < 0. \tag{15}$$

The strict sign is an integer comparison: $2401 < 3328$. Equation (15) is independent of the logarithm convention.

This pair also closes the two related conjectures in the cited literature. First, the proposed spectral inequality compares the maximum global unitary-orbit relative entropy with the minimum marginal unitary-orbit relative entropies. Flat spectra make every rearrangement identical, so its gap is again Eq. (15). Second, replacing U_{AB} by $U_A \otimes U_B$ cannot help because the global first argument in Eq. (12) is invariant under every unitary, including every local one.

For completeness, the full two-qubit family in Eq. (6) has

$$\Delta_{2,2}(\varepsilon) = \frac{1}{4}\log\left[\frac{16\varepsilon(1-2\varepsilon)^4}{1-3\varepsilon}\right], \qquad 0 < \varepsilon < \frac{1}{3}. \tag{16}$$

Hence the rational certificate is not an isolated point.

Corollary 2 (Open sets of violations)**.** *The independent-unitary conjecture and its local-unitary strengthening fail on open sets of full-rank two-qubit state pairs.*

Proof. For the independent-unitary statement, define

$$\begin{aligned}\mathcal{G}(\rho,\sigma) = &\max_{U_{AB}} D(U_{AB}\rho_{AB}U_{AB}^\dagger\|\sigma_{AB})\\ &- \min_{U_A} D(U_A\rho_A U_A^\dagger\|\sigma_A) - \min_{U_B} D(U_B\rho_B U_B^\dagger\|\sigma_B).\end{aligned} \tag{17}$$

The conjecture holds for a pair if and only if $\mathcal{G} \geq 0$. Relative entropy is continuous when its second argument is full rank, and maxima or minima of a continuous function over a compact unitary group depend continuously on the states. Thus $\mathcal{G}$ is continuous. At the pair in Eq. (12), it equals the strictly negative number in Eq. (15); the sign persists in a neighborhood. The same argument applies to the maximum of the locally constrained gap over $U_A \times U_B$. □

IV. DISCUSSION

The counterexample identifies a structural obstruction rather than a noncommutative one. Both states commute, and the maximally mixed first argument removes all unitary-orbit freedom. In this setting, the conjecture would require the geometric-mean determinant relation (5); the family in Eq. (6) violates it by an arbitrarily large relative-entropy gap. Unitary rotations therefore cannot provide a dimension-independent repair of superadditivity for correlated reference states.

The scope is equally clear. The result does not affect data processing, the standard inequality for product references in Eq. (2), or correlation-dependent quasi-factorization bounds. In particular, valid extensions for general reference states introduce a multiplicative factor controlled by the correlations in σ_{AB} [3]. Our family shows why some such correction is unavoidable. A natural remaining question is to determine the optimal factor under unitary-orbit optimization and specified spectral or correlation constraints on the reference state.

[1] L. Zhang *et al.*, Phys. Lett. A **377**, 1794 (2013).
[2] L. Zhang *et al.*, Int. J. Theor. Phys. **53**, 2959 (2014).
[3] A. Capel *et al.*, IEEE Trans. Inf. Theory **64**, 4758 (2018).